

NOTES: CLASS-10

Complex form of Fourier series

Problem 2. Find the complex form of the Fourier Series of $f(x) = \cos ax$ $-\pi \leq x \leq \pi$.

Solution : Here period, $2l = 2\pi \Rightarrow l = \pi$

The complex form of The Fourier Series of $f(x)$ is given by,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/l}$$

where

$$\begin{aligned} c_n &= \frac{1}{2l} \int_{-l}^l f(x) e^{-in\pi x/l} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\pi x} \cos ax dx. \\ &= \frac{1}{2\pi} \left\{ \frac{e^{-inx}}{a^2 + i^2 n^2} [-i \cos ax + a \sin ax] \right\}_{-\pi}^{\pi} \\ &\quad \left\{ \text{Since, } \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx] \right\} \\ &= \frac{1}{2\pi} \left\{ \frac{e^{-in\pi}}{a^2 - n^2} [-i \cos a\pi + a \sin a\pi] - \frac{e^{in\pi}}{a^2 - n^2} [-i \cos a\pi - a \sin a\pi] \right\} \\ &= \frac{(-e^{-in\pi} + e^{-in\pi})}{2\pi(a^2 - n^2)} \cdot i \cos a\pi + \frac{(e^{-in\pi} + e^{-in\pi})}{2\pi(a^2 - n^2)} \cdot a \sin a\pi \\ &= \frac{\sin n\pi}{\pi(a^2 - n^2)} \cdot i \cos a\pi + \frac{\cos n\pi}{\pi(a^2 - n^2)} \cdot a \sin a\pi \\ &\quad \left\{ \text{Since, } \frac{(-e^{-in\pi} + e^{-in\pi})}{2} = \sin n\pi \text{ \& } \frac{(e^{-in\pi} + e^{-in\pi})}{2} = \cos n\pi \right\} \\ &= \frac{\sin n\pi}{\pi(a^2 - n^2)} \cdot i \cos a\pi + \frac{a \cos n\pi}{\pi(a^2 - n^2)} \cdot \sin a\pi \quad \left\{ \text{Since, } \sin n\pi = 0 \text{ \& } \cos n\pi = (-1)^n \right\} \\ \Rightarrow c_n &= \frac{(-1)^n}{\pi(a^2 - n^2)} \cdot a \sin a\pi \end{aligned}$$

Therefore, the complex form of Fourier Series for $f(x)$ is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/l}$$

$$\Rightarrow \cos ax = \sum_{n=-\infty}^{\infty} \frac{(-1)^n a \sin a \pi}{\pi(a^2 - n^2)} \cdot e^{inx}$$

REFERENCES:

1. Higher Engineering Mathematics-B.S. Grewal
2. Higher Engineering Mathematics-B.V. Ramana
